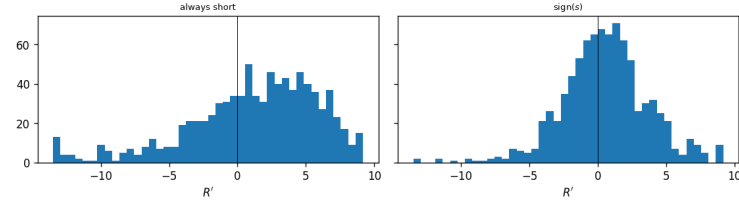


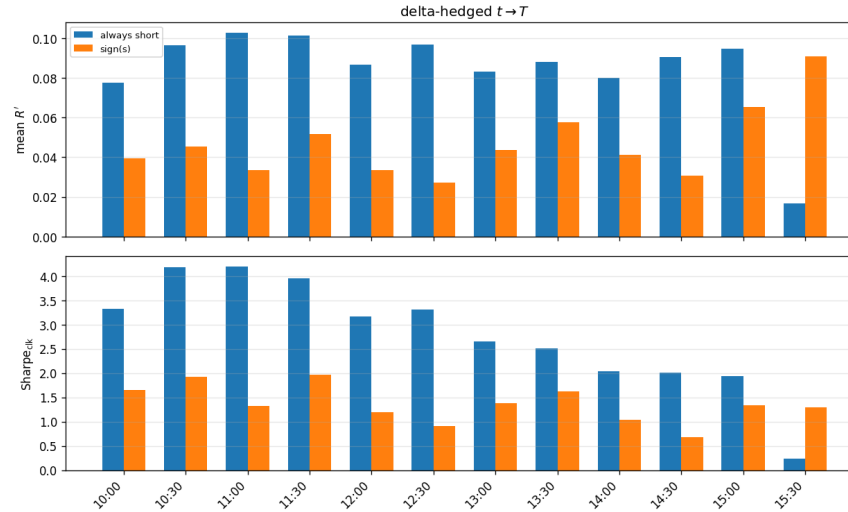
ODTE nearest-OTM straddle: per-rule return summary, pooled over the twelve entry windows 10:00–15:30 ET (overlapping remaining-session books), models compared on the same 866 days

	n	mean	std	min	max	skew	ex. kurt	t	Sharpe _{ann}	n_{buy}	%buy	Sharpe _{ann} ^x
<i>Short volatility (always short)</i>												
all models (no forecast)	866	1.014	5.487	-54.71	10.01	-2.40	14.95	5.44	2.93	0	0.0	2.21
<i>sign(s) (± 1 on the sign of s)</i>												
baseline (HAR + calendar OLS)	866	0.446	3.633	-16.36	48.77	2.45	36.72	3.61	1.95	266	30.7	0.86
block-diagonal ridge	866	0.560	3.727	-20.53	54.71	3.21	52.60	4.42	2.38	252	29.1	1.33
block-diagonal ridge, without the FOMC columns	866	0.532	3.690	-16.36	54.71	3.55	53.83	4.25	2.29	267	30.8	1.22
LightGBM [†]	866	0.536	3.725	-20.53	54.71	3.32	52.85	4.23	2.28	277	32.0	1.23
XGBoost [†]	866	0.524	3.738	-20.53	54.71	3.30	52.17	4.13	2.23	271	31.3	1.17
lasso (causally tuned) [†]	866	0.494	3.658	-20.53	48.77	2.36	36.43	3.98	2.14	276	31.9	1.07
lasso ($\alpha = 10^{-4}$)	866	0.561	3.491	-20.53	37.84	1.00	16.98	4.73	2.55	248	28.6	1.42
elastic net (causally tuned) [†]	866	0.470	3.748	-20.53	54.71	3.12	51.80	3.69	1.99	266	30.7	0.94

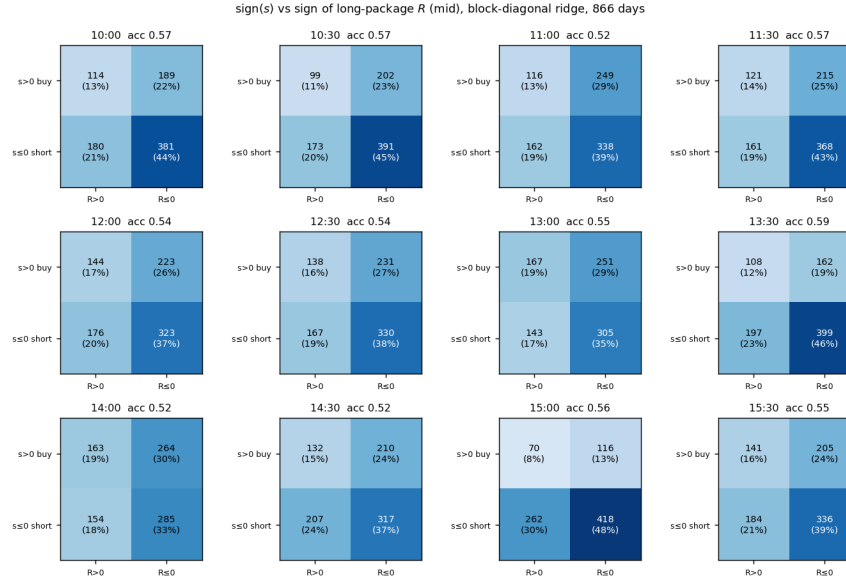
block-diagonal ridge, midpoint fills: daily sum over the twelve entry windows 10:00-15:30 ET; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



Distribution of the return of the daily sum over the twelve entry windows, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).



Mean R' and Sharpe_Δ by entry clock (grouped bars). Sharpe_Δ is that clock's daily series $\times \sqrt{252}$, not the pooled Sharpe_{ann}. Delta-hedged $t \rightarrow T$: hold entry K to official close, rebalance Δ every 30 minutes.



Confusion matrix per entry clock: $\text{sign}(s)$ (buy if $s > 0$) against the sign of the long-package midpoint return. 15:30 is the paper trade.

Each row is the daily sum of the twelve overlapping remaining-session books: enter at each clock 10:00–15:30 ET, hold those strikes to the official close, delta-hedge every 30 minutes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule at the crossed spread: entry at the touch (the ask when long, the bid when short) and cash-settle at the official close (no exit spread). One expiration day = one return; mid fill. Every t here is the plain $t = \sqrt{n} \cdot \text{mean}/\text{std}$, with no autocorrelation correction. “ex. kurt” is the bias-corrected sample excess kurtosis (Gaussian = 0). The short-volatility rule takes no forecast, so it is one row for all models. **Two Sharpes.** Both multiply by $\sqrt{252}$ because an expiration day is the unit of time (same year-length as the paper). They are not the same object. On a *single-clock* page, $\text{Sharpe}_{\text{clk}} = (\text{mean}/\text{std}) \times \sqrt{252}$ of that clock’s series: one remaining-session short per expiration day (enter at this clock, hold those strikes to the close, delta-hedge every 30 minutes). It answers “if I only entered at this clock, what is my annual Sharpe?” It is *not* the book’s $\text{Sharpe}_{\text{ann}}$. On the *pooled* page, $\text{Sharpe}_{\text{ann}} = (\text{mean}/\text{std}) \times \sqrt{252}$ of $R_d^{\text{day}} = \sum_t R'_{d,t}$, the sum of the day’s twelve overlapping remaining-session books. That reserved name is the book’s daily P&L after pooling the session. The twelve books overlap; they are not twelve successive 30-minute holds. $\text{Sharpe}^\times$ is the same convention at the crossed spread. Every other column is daily. n_{buy} is the number of expiration days with position > 0 (always-short is 0; on the pooled page it is days the sum of the twelve positions is positive); %buy is $100 \cdot n_{\text{buy}}/n$, the notebook’s column. The other column this table adds to the deck’s is $\text{Sharpe}_{\text{ann}}^\times$, the same rule’s annualized Sharpe at the *crossed spread* instead of the midpoint; every other column is the deck’s.

Forecasts: baseline (HAR + calendar OLS); block-diagonal ridge (HAR block and exogenous block, separate penalties) on the design carrying the FOMC calendar block, and the same ridge on the earlier design without those columns, as a diagnostic row; LightGBM and XGBoost on the all-features design; lasso on the same design, causally tuned vs. fixed $\alpha = 10^{-4}$; elastic net (causally tuned). † marks a column still fitted on the earlier design panel; a run of those four on the panel of record is pending. The signal is $s_t = \widehat{RV}_t - IV_{\text{hr}}^2 h_t w_t$: the forecast against the implied variance of the same window. w_t is the expanding mean of each prior day’s remaining-session share of realized variance, $RV_t / \sum_{s \geq t} RV_s$ (in-fit panel history back to 2001), not the ratio of trailing clock-means. On the bars whose vendor implied volatility is a censored solver node the deck’s treatment is used here too — the volatility that reproduces the package midpoint, recovered by bisection over the remaining h_t hours — so every bar carries a signal and no bar sits flat.